EE4-48
EE9-CS6-2
EE9-FPN1-03

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE PART IV: MEng and ACGI

POWER SYSTEM DYNAMICS, STABILITY AND CONTROL

Thursday, 13 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : B. Chaudhuri

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The Questions

1.

a)

i) What are the differences between the lightning and switching transients? [3]

ii) How are these transients electrically characterised? [4]

b)

i) What is *resonance stability* in the context of power system operation? [5]

ii) How has the nature of the resonance stability been influenced by the increasing presence of high-power electronics interfaced generation in the power network? [3]

iii) Why is the electromagnetic transient model required for resonance stability study whereas RMS phasor model is adequate for electromechanical stability study? [4]

iv) Describe the purpose of a power system stabiliser (PSS) in an interconnected power system. [4]

ii) List two stabilising feedback signals that can be used in PSS for providing stability in system operation. [2]

[25 marks]

2.

a)

i) The rotor of a synchronous generator is of solid construction whereas the stator is made up of laminated materials. Provide technical reasons for such a constructional practice. [4]

ii) Even though the rotor of a synchronous generator is of solid construction, it is reinforced with additional windings. Describe the purpose of such windings. [4]

iii) How are these additional rotor windings modelled for stability study? [2]

(iv) Why do we ignore the presence/contribution of these windings in steady state analysis of synchronous generator? [2]

iii) Modern synchronous generators are designed to have a large steady state synchronous reactance (between 1.5 to 2.0 pu). However, when a direct terminal short circuit occurs, the initial currents are 4-5 times higher than normal load current. Explain the reason. [5]

iv) Why does a synchronous generator offer higher overload limit while overexcited compared to when it is under excited? [4]

v) What is saliency power in the contexts of synchronous machine operation? [2]

vi) How does it vary with generator load angle (δ)? [2]

[25 marks]

3.

a)

i) What is Park Transformation (d-q) in the contexts synchronous machine dynamics? [3]

ii) Why is Park Transformation preferred over a stationary 3-phase (a-b-c) reference frame? [3]

iii) Despite having represented a synchronous machine in d-q reference frame, why another transformation is required when the generator under analysis is part of an interconnected power system network? [3]

iv) Why is it difficult to implement fast control for hydro turbine? [3]

v) Why is stator transient neglected in transient stability study? [3]

vi) A synchronous generator produces a voltage of 20 kV line to line at the generator terminal bus. This bus is part of a large network which has a reference node at another location. The power flow solution produces an angle (θ) of 5 degrees for this bus voltage. The generator load angle (δ) is 35 degrees. Compute the direct and quadrature axis components of this terminal voltage in generator's reference frame. [4]

b)

i) Despite having no power electronics and cheaper cost why are Type 1 and Type 2 wind turbine technologies not preferred in modern wind farms? [3]

ii) A wind turbine is designed with 4 m/s cut-in speed and 24 m/s cut-out speed. About 400 such turbines are installed in a large offshore wind farm. The wind hits the first row with a speed of 4.2 m/s. It was observed that some of the turbines in the third row counted from the incoming wind direction were not spinning. Explain clearly what could have been the reason(s). [3]

[25 marks]

4.

a)

i) Why are gearboxes necessary in all of first three types of wind turbine generator? [2]

iii) How does one model the gear box for stability studies? [2]

iv) Why does Type 4 wind turbine usually have large number of poles? [3]

b)

i) How does the *stall* differ from the *pitch* in the contexts of wind power generation control? [3]

ii) How is stall control implemented? [3]

iii) Why is the stall control mechanism avoided in all modern wind turbines? [3]

iv) Why is the generator torque/power control preferred over the pitch control at the sub-rated wind condition? [3]

c)

(i) Why is the *transient stability* less of concern to the system operators with significant wind generation? [3]

(ii) How can wind generators be operated to provide the *frequency stability* in power system operation? [3]

[25 marks]